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A case study:
OOS results in appearance test at OMCL
leading to the revision of specifications
from the manufacturer side

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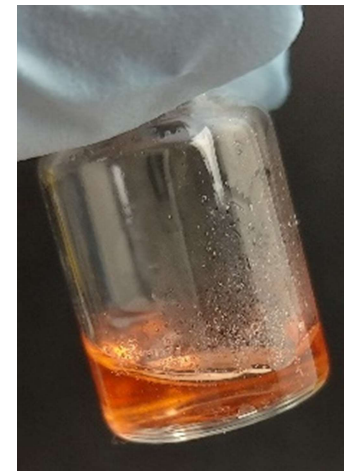
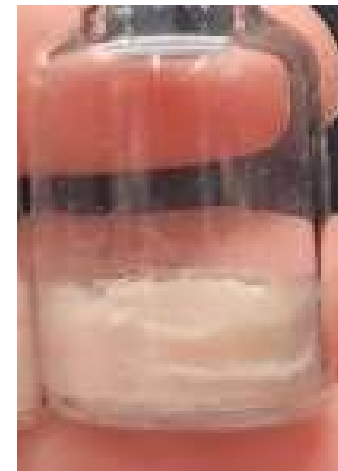
Priorix-Tetra: product information

- Priorix-Tetra is a multicomponent vaccine against Measles, Mumps, Rubella and Varicella
- Vaccine is licensed since 2005
- Presentation: a vial with lyophilized vaccine, reconstituted with WFI



Priorix-Tetra: specification for appearance

“Whitish to slightly pink coloured cake or powder contained in a glass vial sealed with a rubber stopper. After reconstitution with the diluent: clear peach to fuchsia pink coloured solution”



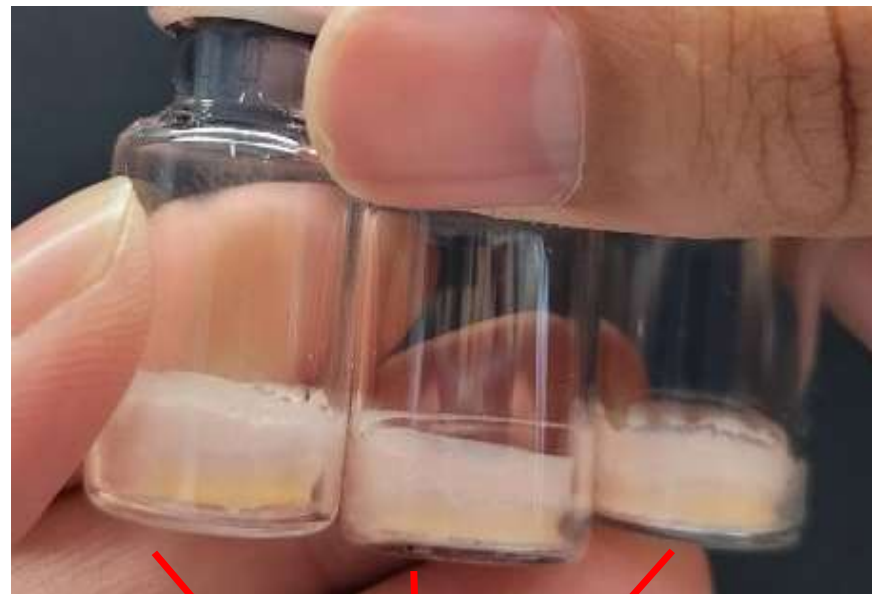
OOS results for appearance testing at OMCL

- Yellowish and layered cakes were observed in some vials of Priorix-Tetra



OOS vials

Valid vial



OOS vials



OOS vial

- Undissolved white particles have also been found after reconstitution in some vials

Impacted Priorix-Tetra batches

Sciensano

Retest of retained samples (8 different lots), by 3 different operators in total. For each of the 8 lots, yellowish part of the cake were observed in some vials.

The white particles after reconstitution have been seen in 3 lots in total (more difficult to observe).

PEI

Retest of all batches from 2024 by two different operators:

All four batches from 2024 showed a yellowish and layered cake in retest of visual inspection. (retest of three vials/batch)

No white particles were observed after reconstitution.

Manufacturer's investigation

- Yellow colour of the cake is linked to changes in light absorption and reflection of a retracted cake – density of the environment around the phenol red affects how light interacts with it (transition from light pink to yellow/orange)
- Stability study in 2016 demonstrated the non-product impact of the yellowish colour but at that time, regulators had not been notified about this yellow ring at the bottom of the cakes
- The white particles (cell debris) are endogenous and as such their presence is inherent to the manufacturing process of a varicella containing vaccine. Those particles have been characterised, and the study confirmed their MRC-5 cell origin

Manufacturer's investigation

- There was **no change in manufacturing processes** that could significantly affect product appearance
- Visual inspection and **QC SOPs at the manufacturer** considered **as acceptable** this kind of change of colour of the cake and the presence of endogenous white particles after reconstitution of the Priorix-Tetra product
 - these vials were not classified as OOS and therefor not discarded during visual inspection process at GSK

Manufacturer's conclusion

- The investigation confirmed that the **yellowish cake** aspect before reconstitution and the **presence of endogenous white particles** after reconstitution are **normal characteristics of the Priorix-Tetra product**. They are inherent to the nature of the product and do not impact the quality, potency and safety of the product.
- **Registered product specifications need to be updated** in order to align with the product's inherent properties, as reflected in current QC testing and visual inspection procedures at GSK
- GSK sent to PEI & Sciensano a statement explaining this situation (beginning 2025)

OMCL decision

- Until the variation of specification is approved, Sciensano and PEI agreed on a **joint interim release strategy**
 - Issue only national certificates
 - No EU certificates are released
 - WHO certificate states that the appearance test is not conform to the current specification described in the regulatory dossier

New appearance specification

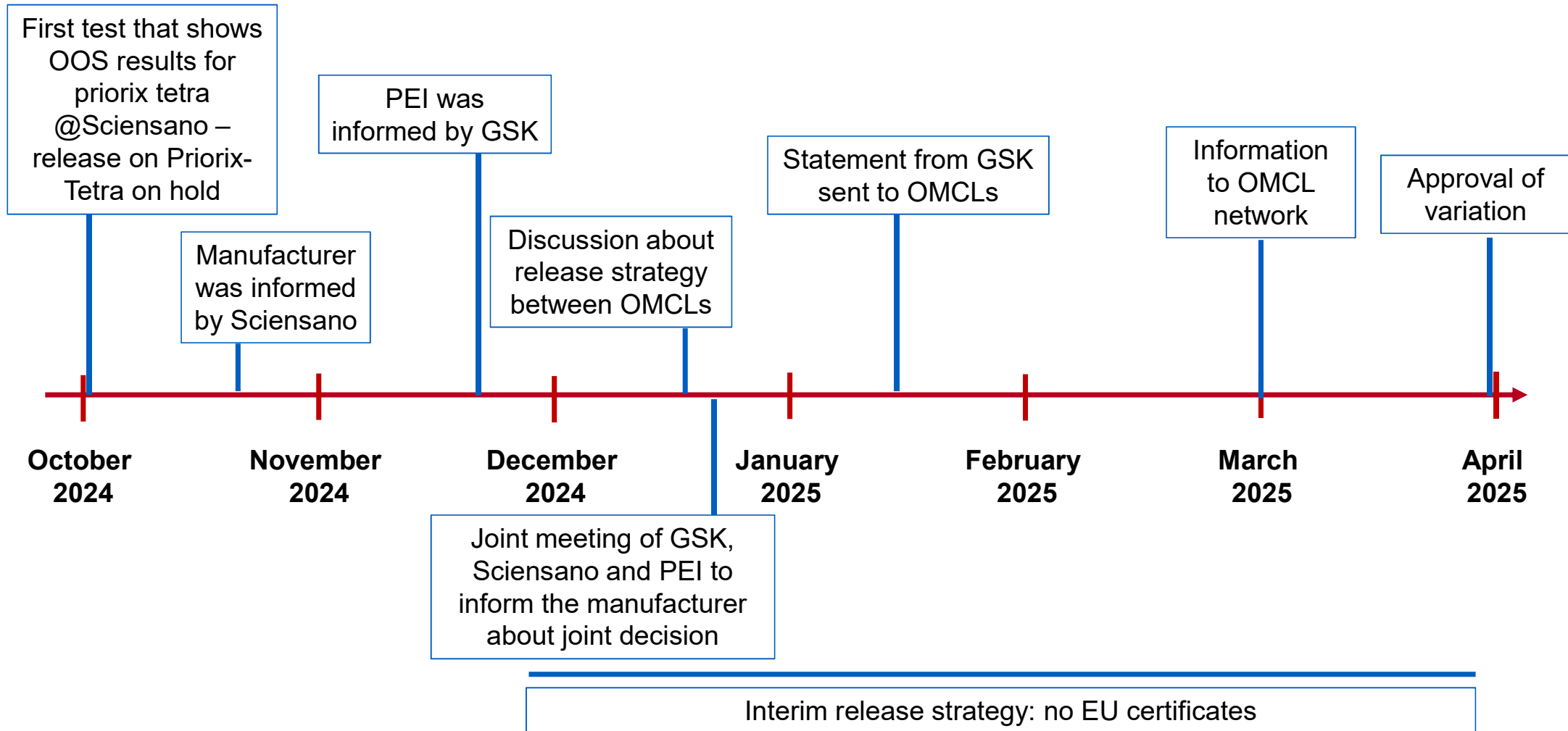
- *March 29: Approval of variation*

“Whitish to slightly pink coloured cake or powder, which may be partially yellowish and contained in a glass vial sealed with a rubber stopper.

After reconstitution with the diluent:

clear peach to fuchsia pink coloured solution. Presence of endogenous translucent particles may be observed”

Timeline





sciensano

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THANK YOU VERY MUCH
FOR YOUR ATTENTION

Back-up slides

Tests performed at Sciensano (release + investigation tests)

Lot	Test date	Operator	# Analyzed samples	Results BEFORE reconstitution		Results AFTER reconstitution	
				# OK	# NOK	# OK	# NOK
AMRVA745A	01-10-24	Operator A	3	3	0	0	3
AMRVA745A	17-10-24	Operator B	10	7	3	10	0
AMRVA745A	22-10-24	Operator C	10	6	4	10	0
AMRVA704A			3	0	3	3	0
AMRVA703A			3	0	3	3	0
AMRVA702A			3	1	2	3	0
AMRVA722A	07-11-24	Operator B	17	10	7	NA	
AMRVA703A			16	3	13		
AMRVA702A			16	1	15		
AMRVA704A			20	2	18		
AMRVA665A			12	1	11		
AMRVA728A			10	5	5	7	3
AMRVA705A			10	2	8	7	3

A6

Flore Laurent : to remind you, after this investigation, 3 lots have been considered OOS (the one highlighted in orange - AMRVA745A is the lot that triggers all this problematic. The other lots, in white, were already released and had been considered valid during a first release test, that was conducted before the 01/10/24.

Autor; 23.04.2025

Back-up slide

complete explanation given by GSK regarding the retraction and coloration mechanism of the cake

1. Retraction phenomenon:

- **Physical retraction occurs during lyophilization** if product temperature is near $T_g'^{(1)}$.
- Visible at the bottom of the cake (shrunk aspect) with or without a yellowish hue

2. Color Change:

- Density of the environment around the phenol red affects how light interacts with it. If environment becomes denser (due to retraction), it induces **changes in light absorption and reflection**
- Transition from light pink to yellow/orange

3. Conclusion:

- A retraction without yellowish aspect can be observed
- A yellowish aspect without retraction is not observed.

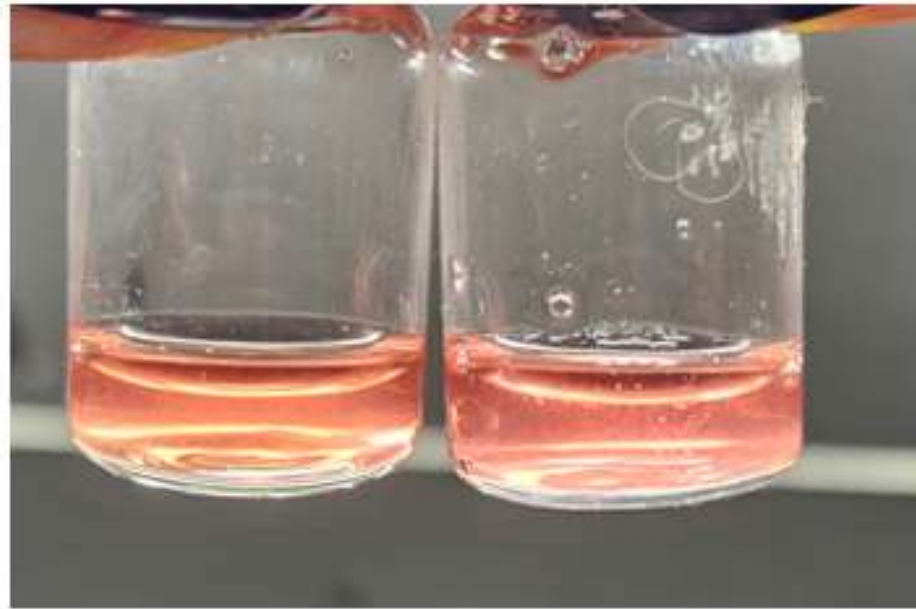
Back-up slide

complete explanation given by GSK regarding the white particules

- The presence of white particles observed in the reconstituted Priorix Tetra vaccine is a normal characteristic of the product. These particles are endogenous and inherent to the manufacturing process of varicella containing vaccines
 - As Varicella Zoster Virus is highly associated to host cell membranes
 - and as GSK varicella manufacturing process does not contain any filtration step (only a clarification by centrifugation),
 - ↳ host cell (MRC-5) / host cell residuals are not completely removed from the product
- The visible particles observed in reconstituted Priorix Tetra Drug Product have been characterised¹ in order to identify, characterise and classify them as «*inherent*» and «*normal to observe within the considered product*» from «*non-inherent*» particles (abnormal to observe). This study has been conducted according to requirements of European and US pharmacopeia chapters related to visible particles^{2 3}
 - ↳ The study confirmed the MRC-5 cell origin and the endogenous characteristic of these visible particles
 - ↳ These visible particles are considered as being inherent to the varicella containing vaccine manufacturing process and therefore a normal characteristic of the product.
- At the Final Container step, the « *Particulate matter by Visual Inspection* » test is performed (In-process Quality Decision test) and addresses the absence of extraneous visible particles (acceptance criterion: « *Essentially free from extraneous visible particulates after reconstitution with water for injection* »)

Back-up slide

Picture of white particles found during appearance test at Sciensano

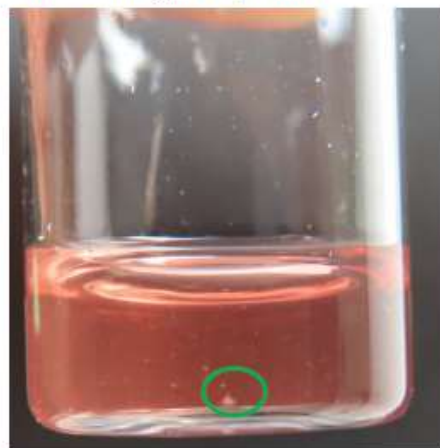


ANNEXE : 05 CAS PARTICULIER DE VARICELLE (VAR) ET PRIORIX TETRA (MRV)

Back-up slide

Exemple de débris cellulaires Priorix Tetra

AMRVA290A grandes particules



AMRVA363AA petites particules



AMRVA461A particule allongée



DESCRIPTION DES VACCINS